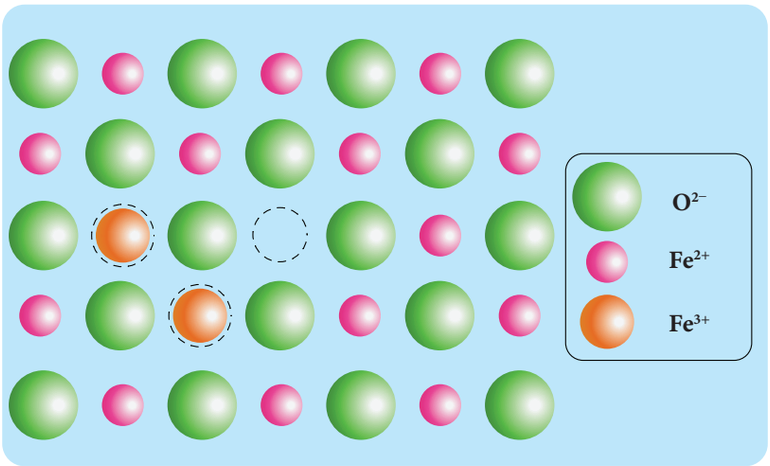
ZnO is colourless at room temperature. When it is heated, it becomes yellow in colour. On heating, it loses oxygen and thereby forming free Zn^{2+} ions. The excess Zn^{2+} ions move to interstitial sites and the electrons also occupy the interstitial positions.

6.7.4 Metal deficiency defect:

Metal deficiency defect arises due to the presence of less number of cations than the anions. This defect is observed in a crystal in which, the cations have variable oxidation states.



Metal Deficiency Defect

For example, in FeO crystal, some of the Fe^{2+} ions are missing from the crystal lattice. To maintain the electrical neutrality, twice the number of other Fe^{2+} ions in the crystal is oxidized to Fe^{3+} ions. In such cases, overall number of Fe^{2+} and Fe^{3+} ions is less than the O^{2-} ions. It was experimentally found that the general formula of ferrous oxide is Fe_xO, where x ranges from 0.93 to 0.98.

6.7.5 Impurity defect:

A general method of introducing defects in ionic solids is by adding impurity ions. If the impurity ions are in different valance state from that of host, vacancies are created in the crystal lattice of the host. For example, addition of $CdCl_2$ to AgCl yields solid solutions where the divalent cation Cd^{2+} occupies the position of Ag^+. This will disturb the electrical neutrality of the crystal. In order to maintain the same, proportional number of Ag^+ ions leaves the lattice. This produces a cation vacancy in the lattice, such kind of crystal defects are called impurity defects.



Energy harvesting by piezoelectric crystals:

Piezoelectricity (also called the piezoelectric effect) is the appearance of an electrical potential across the sides of a crystal when you subject it to mechanical stress. The word piezoelectricity means electricity resulting from pressure and latent heat. Even the inverse is possible which is known as inverse piezoelectric effect.

If you can make a little amount of electricity by pressing one piezoelectric crystal once, could youmake a significant amount by pressing many crystals over and over again? What happens if we bury piezoelectric crystals under streets to capture energy as vehicles pass by?

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